

Distributivity and the Commensurability of Value-Definite Realism and Empirical Adequacy

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Abstract

When does an empirically adequate theory admit a value-definite completion — an assignment of simultaneous definite values to all observables? We show that the answer depends on the algebraic structure of the observations. For a directed system of Boolean algebras carrying compatible charges, the Stone construction produces an unconditional measure on the dual space, and every observable admits a simultaneous definite value via ultrafilters. The passage to a probability measure on an underlying state space requires only σ -additivity, and the choice of state space is unconstrained by the algebra [?].

For orthomodular lattices — the algebraic framework for incompatible observations — the situation is structurally different. The McDonald–Bimbó dual space carries principal filters as distinguished structure, and the Kochen–Specker theorem blocks global value-definiteness. Probabilistic realism (descent to a probability measure) remains available via Gleason-type results, but value-definite realism does not.

The dividing line is distributivity. We identify a four-level coherence hierarchy — consistency, contextual coherence, probabilistic coherence, value-definite coherence — and show that distributivity of the observation algebra is the exact condition under which all four levels are mutually commensurable.

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1 Introduction

A central question in the philosophy of science asks when an empirically successful theory can be completed to a realist one — one that assigns definite properties to an underlying reality, not merely adequate predictions to observable phenomena. Van Fraassen [?] argues that science need only aim at *empirical adequacy*: agreement with all observable phenomena. The scientific realist demands more — an underlying state space whose properties generate the observations.

This debate is usually conducted in philosophical terms. We give it a mathematical formulation and show that its resolution depends on the algebraic structure of the observations.

Three kinds of realism

We distinguish three nested positions, each a strengthening of the previous:

- (i) **Empirical adequacy (EA).** A measure on the dual space of the observation algebra that agrees with all observed charges. No commitment to an underlying state space.
- (ii) **Probabilistic realism (PR).** A probability measure on an underlying state space $(\Omega, \sigma(\mathcal{C}))$ that generates the observations. Requires descent from the dual-space measure.

- (iii) **Value-definite realism (VDR).** Every observable simultaneously has a definite value. A global dispersion-free state — equivalently, a 2-valued homomorphism on the full observation algebra.

These are nested: VDR implies PR implies EA. The question is which implications reverse, and what governs the answer.

The role of the observation algebra

Consider two settings:

Compatible observations. When all observations can be performed simultaneously, the observation algebra is a Boolean algebra — a complemented distributive lattice. Every Boolean algebra admits 2-valued homomorphisms (ultrafilters exist by Zorn’s lemma). In this setting, all three positions are available and mutually commensurable. The passage from EA to PR requires σ -additivity of the charges [?]; the further passage to VDR is free.

Incompatible observations. When some observations cannot be performed simultaneously — as in quantum mechanics, contextual experimental design, or distributed systems with interfering sensors — the observation algebra is an orthomodular lattice (OML), a complemented but non-distributive lattice. The Kochen–Specker theorem [?] shows that for the prototypical case $L(H)$ with $\dim H \geq 3$, no 2-valued homomorphism exists: VDR is blocked. Probabilistic realism remains available via Gleason’s theorem [?], and empirical adequacy is unconditional.

The dividing line is distributivity.

Contributions

We develop the following:

1. A unified framework based on directed systems of complemented lattices, with the Boolean and orthomodular cases as instances (§??–§??).
2. A four-level coherence hierarchy — consistency, contextual coherence, probabilistic coherence, value-definite coherence — with the obstruction at each level identified (§??).
3. A structural theorem: distributivity of the observation algebra is the exact condition under which all four levels are mutually commensurable (§??).
4. An analysis of the philosophical consequences: the realism/empiricism debate has different mathematical answers depending on the observation algebra (§??).

2 The Boolean case

We recall the framework and results of [?], which treats directed systems of Boolean algebras.

Directed systems and compatible charges

Let $(\iota, \leq)$ be a directed set indexing a family of Boolean algebras $\{B_i\}_{i \in \iota}$ with injective homomorphisms $\varphi_{ij} : B_i \rightarrow B_j$ for $i \leq j$. Write $\mathcal{C} = \bigcup_i B_i$ for the direct limit — the *cylinder algebra*.

A *compatible family of normalised charges* is a collection $\{\ell_i\}$ with $\ell_i : B_i \rightarrow [0, 1]$ finitely additive, $\ell_i(1) = 1$, and $\ell_j \circ \varphi_{ij} = \ell_i$ for $i \leq j$.

The Stone construction

The Stone space $\text{St}(\mathcal{C})$ is the space of ultrafilters on $\mathcal{C}$, compact and totally disconnected. Each $\omega \in \Omega$ determines a principal ultrafilter $\text{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$.

Any compatible family of normalised charges extends to a σ -additive Baire probability measure $\hat{\mu}$ on $\text{St}(\mathcal{C})$ — unconditionally, with no assumption of σ -additivity [?, ?]. This is the *empirically adequate* object: it agrees with all charges on cylinder events.

Descent and σ -additivity

The measure $\hat{\mu}$ lives on $\text{St}(\mathcal{C})$, not on Ω . Descent to a probability measure P on $(\Omega, \sigma(\mathcal{C}))$ requires $\hat{\mu}(\text{pure}(\Omega)) = 1$ — that $\hat{\mu}$ concentrates on the principal ultrafilters. This holds if and only if each charge ℓ_i is σ -additive [?].

Value-definiteness in the Boolean case

Every Boolean algebra admits ultrafilters (by Zorn’s lemma), and every ultrafilter is a 2-valued homomorphism. A 2-valued homomorphism assigns to each element $a \in \mathcal{C}$ a definite value in $\{0, 1\}$, respecting the lattice operations.

Moreover, any subset $S \subseteq \text{St}(\mathcal{C})$ that projects surjectively onto each $\text{St}(B_i)$ can serve as a realisation space: set $\Omega = S$ with evaluation maps given by the restriction maps. The algebra imposes no constraint on the choice of Ω [?].

Proposition 2.1 (Boolean commensurability). *For a directed system of Boolean algebras with compatible charges:*

- (a) *EA is unconditional: $\hat{\mu}$ on $\text{St}(\mathcal{C})$ always exists.*
- (b) *PR holds iff each ℓ_i is σ -additive.*
- (c) *VDR holds unconditionally: $\text{St}(\mathcal{C})$ consists entirely of 2-valued homomorphisms.*

In particular, EA, PR, and VDR are mutually commensurable: the only obstruction is σ -additivity at the EA–PR transition, and it is a condition on the charges, not the algebra.

Proof. Part (a) is the Stone construction [?, ?]. Part (b) is [?, Theorem 1]. Part (c): ultrafilters on a Boolean algebra are exactly the 2-valued homomorphisms; their existence follows from Zorn’s lemma applied to the filter generated by any non-zero element. $\square$

3 Orthomodular observation algebras

The Boolean assumption is that all observations are compatible: any finite collection can be performed simultaneously. When this fails — when some observations preclude others — the observation algebra loses distributivity. The appropriate generalisation is the orthomodular lattice.

Orthomodular lattices

Definition 3.1. An *ortholattice* is a bounded lattice $A = (A, \wedge, \vee, {}^\perp, 0, 1)$ with an orthocomplementation ${}^\perp : A \rightarrow A$ satisfying $a \wedge a^\perp = 0$, $a \vee a^\perp = 1$, $a^{\perp\perp} = a$, and $a \leq b \Rightarrow b^\perp \leq a^\perp$.

Definition 3.2. An *orthomodular lattice* (OML) is an ortholattice satisfying the *orthomodular law*:

$$a \leq b \implies b = a \vee (a^\perp \wedge b).$$

Every Boolean algebra is an OML (the orthomodular law follows from distributivity), but the converse fails. An OML is Boolean if and only if it is distributive [?, Proposition 2.3].

Example 3.3. Let H be a separable Hilbert space with $\dim H \geq 3$. The lattice $L(H)$ of closed linear subspaces, ordered by inclusion with $U^\perp = \{x \in H : \langle x, y \rangle = 0 \ \forall y \in U\}$, is an orthomodular lattice that is not distributive [?].

Example 3.4 (Incompatible experimental design). Consider an experiment with three binary measurements A, B, C , where (A, B) and (A, C) can be performed jointly but (B, C) cannot. The jointly measurable pairs generate Boolean subalgebras, but the full observation algebra is orthomodular, not Boolean. This situation arises in contextual behavioural science, clinical trials with mutually exclusive protocols, and sensor networks with interfering instruments.

Commutativity and classical contexts

Two elements a, b in an OML are said to *commute* if they generate a Boolean subalgebra. A maximal set of pairwise commuting elements forms a *Boolean context* — a maximal Boolean subalgebra of A . Within each context, classical logic and classical probability apply. The non-classical character of the OML arises from the relationships *between* contexts.

States on OMLs

Definition 3.5. A *state* on an OML A is a function $s : A \rightarrow [0, 1]$ with $s(1) = 1$ and

$$a \perp b \implies s(a \vee b) = s(a) + s(b),$$

where $a \perp b$ means $a \leq b^\perp$.

A state is additive on *orthogonal* pairs only — not on arbitrary joins. This is the OML analogue of a finitely additive charge on a Boolean algebra.

Definition 3.6. A state s is *dispersion-free* if $s(a) \in \{0, 1\}$ for all $a \in A$.

A dispersion-free state is a 2-valued homomorphism: it assigns a definite truth value to every proposition in A . On a Boolean algebra, dispersion-free states are exactly the ultrafilters.

The Kochen–Specker obstruction

Theorem 3.7 (Kochen–Specker [?]). *If $\dim H \geq 3$, there is no dispersion-free state on $L(H)$.*

No assignment of simultaneous definite values to all observables is consistent with the algebraic structure. This is not a limitation of knowledge but a structural impossibility: the OML $L(H)$ does not admit a 2-valued homomorphism.

The obstruction is dimensional: for $\dim H = 2$, dispersion-free states exist (every maximal Boolean subalgebra of $L(\mathbb{C}^2)$ is a four-element Boolean algebra, and 2-valued homomorphisms are plentiful). The critical threshold is $\dim H = 3$.

Gleason’s theorem

Despite the absence of dispersion-free states, *probabilistic* states on $L(H)$ are well-behaved:

Theorem 3.8 (Gleason [?]). *If $\dim H \geq 3$, every σ -additive state on $L(H)$ is of the form $s(P) = \text{tr}(\rho P)$ for a unique density operator ρ on H .*

Gleason’s theorem says that probabilistic realism is available for $L(H)$: every state extends to a well-defined probability assignment via the Born rule. What fails is value-definite realism — the demand that every observable have a simultaneous definite value.

4 The OML dual space

In the Boolean case, Stone duality provides the dual space $\text{St}(\mathcal{C})$ of ultrafilters. For OMLs, the analogue is the McDonald–Bimbó duality [?], which we now describe.

Filters on an OML

Let A be an OML. A *filter* on A is a non-empty subset $x \subseteq A$ that is upward closed ($a \in x$ and $a \leq b$ implies $b \in x$) and closed under finite meets ($a, b \in x$ implies $a \wedge b \in x$). A filter is *proper* if $0 \notin x$; the unique *improper* filter is $\omega = A$.

For each $a \in A$, the *principal filter* generated by a is $\uparrow a = \{b \in A : a \leq b\}$. Write $F(A)$ for the collection of all filters on A , and $P(A) \subseteq F(A)$ for the subcollection of principal filters.

The dual space

Definition 4.1 (McDonald–Bimbó [?]). Let A be an OML. The *dual space* of A is the ordered relational topological space

$$S_0(A) = (F(A), \subseteq, \perp_A, P(A), \mathcal{T}),$$

where:

- (i) $F(A)$ is the set of all filters on A ;
- (ii) $\subseteq$ is set inclusion;
- (iii) $x \perp_A y$ iff there exists $a \in A$ with $a \in x$ and $a^\perp \in y$;
- (iv) $P(A)$ is the collection of principal filters;
- (v) $\mathcal{T}$ is the topology generated by the subbasis $\{h(a) : a \in A\} \cup \{F(A) \setminus h(a) : a \in A\}$, where $h(a) = \{x \in F(A) : a \in x\}$.

Theorem 4.2 (McDonald–Bimbó [?]). $S_0(A)$ is an orthomodular space (compact, totally disconnected). The category of OMLs and homomorphisms is dually equivalent to the category of orthomodular spaces and continuous weak p -morphisms. Moreover, $A \cong \text{CO}(S_0(A))^\dagger$, the OML of clopen $\perp$ -stable subsets of $S_0(A)$.

Structural comparison with Stone duality

The critical differences are collected in the following table:

Feature	Boolean (Stone)	OML (McDonald–Bimbó)
Dual points	Ultrafilters	All filters
Distinguished subset	None	$P(A)$ (principal filters)
Clopens	Boolean algebra	OML
Orthogonality	Trivial	Non-trivial ($\perp_A$)
2-valued homomorphisms	Plentiful (Zorn)	May not exist (KS)

In Stone duality, the principal filters (i.e., the principal ultrafilters $\text{pure}(\omega)$ for $\omega \in \Omega$) are *invisible*: they play no distinguished role in the dual space. Which ultrafilters are principal depends on the choice of Ω [?].

In the McDonald–Bimbó duality, $P(A)$ is *part of the structure* of the dual space: the orthomodular frame conditions (Definition ??) explicitly reference $P(A)$. This is the algebraic formalisation of the observation that, for non-distributive algebras, the distinction between realised and unrealised observation-patterns is not freely chosen but is constrained by the algebra.

5 Three kinds of realism, revisited

We now have the framework to state the three positions precisely in both settings.

Definition 5.1. Let A be a complemented lattice (Boolean or orthomodular) with a state $s : A \rightarrow [0, 1]$.

- (i) **Empirical adequacy (EA):** s defines a consistent assignment $\mu(h(a)) = s(a)$ on the clopens of the dual space.
- (ii) **Probabilistic realism (PR):** μ extends to a σ -additive probability measure on the Baire σ -algebra of the dual space, concentrating on $P(A)$.
- (iii) **Value-definite realism (VDR):** There exists a dispersion-free state on A — a 2-valued homomorphism assigning simultaneous definite values to all elements.

Proposition 5.2 (OML status). *For $A = L(H)$ with $\dim H \geq 3$:*

- (a) *EA holds: every state defines a consistent assignment on $\text{CO}(S_0(A))^\dagger$.*
- (b) *PR holds: Gleason's theorem (??) gives a σ -additive measure via $s(P) = \text{tr}(\rho P)$.*
- (c) *VDR fails: the Kochen–Specker theorem (??) shows no dispersion-free state exists.*

Comparing with Proposition ??:

Position	Boolean	OML ($\dim \geq 3$)
EA	✓ (unconditional)	✓ (unconditional)
PR	✓ (iff σ -additive)	✓ (Gleason)
VDR	✓ (ultrafilters)	× (Kochen–Specker)

The single structural difference — distributivity versus non-distributivity — controls exactly the VDR row.